PROJECT SYNOPSIS

Cortex Hub

Enterprise AI Component Store & Customization Fabric

A multi-tenant SaaS marketplace for discovering, testing, customizing, licensing, and deploying production-ready AI components.

Project Type	Web-based Enterprise AI Marketplace and SaaS Platform
Primary Domain	Artificial Intelligence, Machine Learning, SaaS, Enterprise Software
Repository	Goldypahal/Cortex-Hub
Database	**PostgreSQL 18** with Prisma ORM; relational persistence for platform data
Frontend	HTML5, modular JavaScript ES Modules, custom CSS
Backend	Node.js, Express REST API, Prisma Client
Prepared On	16 September 2026

Submitted By: _____

Institution / Department: _____

1. Abstract

Cortex Hub is an enterprise-focused AI component marketplace and customization platform. It enables organizations to discover pre-built AI engines, test them through an interactive playground, configure industry-specific requirements, purchase licenses, and track customization and deployment activities. The platform combines a customer-facing marketplace, an internal administration workspace, and an AI services layer supporting retrieval, model routing, citations, usage tracking, and tenant isolation.

A major implementation detail is the use of a real PostgreSQL database rather than a static or browser-only data store. PostgreSQL 18, accessed through Prisma ORM, is used to persist tenants, users, components, licenses, customization requests, support tickets, API keys, usage records, and other transactional entities. This gives the system durable storage, relational integrity, query capability, and a foundation for multi-tenant SaaS operations.

2. Introduction

Organizations increasingly need AI capabilities such as document question answering, customer support automation, resume matching, legal analysis, financial analysis, and workflow orchestration. However, developing these systems independently requires substantial time, technical expertise, infrastructure, and maintenance. Generic chatbot interfaces also often lack reliable citations, access control, cost visibility, and deployment flexibility.

Cortex Hub addresses this gap by presenting AI functionality as configurable, production-oriented components. Customers can evaluate a component before purchase and select deployment, authentication, data-source, branding, and feature requirements through a guided configurator.

3. Problem Statement

• Businesses often lack the resources to build and maintain specialized AI systems from scratch.

• Generic AI tools may not provide reliable source grounding, usage monitoring, or enterprise access controls.

• Customers need to compare components, estimate customization cost, and understand deployment requirements before purchasing.

• AI marketplace operations require persistent records for customers, licenses, invoices, requests, support, and deployment status.

• A scalable relational database is necessary to maintain consistency across multi-tenant business transactions.

4. Objectives

• Develop a centralized marketplace for production-ready AI components.

• Provide interactive demonstrations and a temporary document playground.

• Allow customers to configure AI components according to business and technical requirements.

• Persist platform data using PostgreSQL 18 and Prisma ORM.

• Implement customer and administrator dashboards with role-based access.

• Support licensing, checkout simulation, invoices, API-key management, and usage tracking.

• Provide a foundation for secure, tenant-isolated AI services and future deployment automation.

5. Scope of the Project

Included: public component catalog, search and filters, live sandbox, architecture/BOM configurator, customization workflow, checkout and licensing, customer portal, admin operations dashboard, PostgreSQL-backed persistence, REST API, and AI-service integration points.

Excluded from Version 1: fully automated customization without human review, third-party developer publishing marketplace, native mobile applications, and a complete global multi-currency tax engine.

6. Proposed System

The proposed system is organized into three interconnected planes:

• **Customer-facing plane:** marketplace catalog, product detail pages, playground, configurator, checkout, licensing, and customer portal.

• **Administration plane:** catalog management, customization Kanban, customer management, billing analytics, support, audit records, and deployment tracking.

• **AI services plane:** document ingestion, hybrid retrieval, model selection, citation generation, evaluation, usage/cost logging, and API exposure.

PostgreSQL acts as the system of record for the transactional and operational data shared by these planes. Prisma provides typed database access and schema management between the Express backend and PostgreSQL.

7. Major Modules

Module	Main Responsibilities
Marketplace and Catalog	Browse, search, filter, view, version, publish, and manage AI components.
Live AI Sandbox	Upload temporary PDF/DOCX files, test queries, display citations, cost, token usage, and latency.
Customization Configurator	Capture domain, data sources, authentication, model, deployment, branding, and feature choices.
BOM and Estimation	Calculate estimated implementation cost in USD/INR and expected delivery timeline.
Checkout and Licensing	Simulate payment, calculate GST/VAT, create invoices, and issue license records.
Customer Portal	Manage organization data, API keys, purchases, usage, documentation, support, and requests.
Admin Operations	Manage catalog, users, requests, licenses, revenue, deployments, support, and audit activity.
AI Services Layer	Support ingestion, retrieval, model routing, citations, evaluation, and usage tracking.
PostgreSQL Data Layer	Persist tenants, users, components, licenses, requests, tickets, keys, usage, and related records.

8. Database and Data Persistence

Cortex Hub uses a real PostgreSQL 18 database as its persistent relational data store. The database is not merely a planned integration: the repository documents PostgreSQL configuration, Prisma Client usage, schema pushing, and database seeding commands.

• **Database engine:** PostgreSQL 18.

• **Data-access layer:** Prisma ORM and @prisma/client.

• **Connection:** configured through the DATABASE_URL environment variable.

• **Schema management:** npm run db:push.

• **Initial data:** npm run db:seed, including ten AI engines and a default tenant.

• **Core entities:** tenants, users, components, component versions, pricing/licensing records, customization requests, support tickets, API keys, usage records, and deployment-related data.

• **Benefits:** durable storage, relational constraints, transactions, indexing, structured querying, backup support, and a foundation for row-level tenant isolation.

Example database configuration: DATABASE_URL="postgresql://postgres:@localhost:5432/cortex_hub". PostgreSQL may be hosted locally or through a managed provider such as Supabase or Neon.

9. Technology Stack

Layer	Technology / Implementation
Frontend	HTML5, vanilla JavaScript, ES modules, custom CSS, responsive dark-mode design system
Backend	Node.js, Express, REST APIs, CORS, dotenv
Database	PostgreSQL 18
ORM	Prisma ORM and @prisma/client
AI / Retrieval	Hybrid pgvector dense retrieval and BM25 lexical retrieval; model/provider integration points
Architecture	Component-driven MVC structure using js/views, js/data, js/utils, and server directories
Development	npm, environment variables, database schema push, seed scripts, concurrent frontend/backend execution

10. Methodology

The project follows an iterative full-stack development methodology:

• Requirement analysis using the project SRS and functional requirement groups FR-1 to FR-7.

• Design of the customer, administration, AI-service, and database layers.

• Implementation of frontend views and modular JavaScript components.

• Implementation of Express REST services and Prisma-based PostgreSQL persistence.

• Database schema synchronization and seeding for initial marketplace data.

• Integration of playground, catalog, configurator, licensing, customer, and admin workflows.

• Testing of functional flows, data validation, authorization, and database operations.

• Future extension toward production AI services, managed deployments, and stronger tenant isolation.

11. Functional Requirements Summary

• The system shall display published AI components with descriptions, categories, pricing, versions, and architecture details.

• The system shall support search, filters, live demos, and temporary playground sessions.

• The system shall accept configuration choices and create customization requests.

• The system shall calculate estimates and support checkout, invoices, and license records.

• The system shall allow customers to manage API keys, usage, purchases, documentation, and support tickets.

• The system shall provide administrator role-based access, catalog CRUD, request tracking, revenue analytics, and deployment tracking.

• The system shall persist platform data in PostgreSQL through Prisma ORM.

• The system shall support AI-service operations such as ingestion, retrieval, citations, model selection, and usage logging.

12. Non-Functional Requirements

• **Security:** authentication, authorization, API-key protection, tenant isolation, and audit logging.

• **Performance:** responsive interfaces, efficient database queries, and measurable AI latency/cost telemetry.

• **Scalability:** modular backend services and PostgreSQL-based persistence suitable for cloud deployment.

• **Reliability:** transactional consistency for licensing, requests, and customer records.

• **Maintainability:** modular JavaScript, MVC organization, Prisma schema, and documented environment setup.

• **Privacy:** temporary playground data, configurable retention, and compliance-oriented design.

13. Expected Outcomes

• A working enterprise AI marketplace interface with ten predefined AI component offerings.

• A PostgreSQL-backed application capable of retaining and querying business data across sessions.

• A guided customization and estimation workflow for business customers.

• Customer and administrator workspaces for operational management.

• A foundation for integrating real AI pipelines, model providers, APIs, SDKs, and deployment infrastructure.

14. Future Enhancements

• Complete production AI execution services with real model routing and retrieval pipelines.

• Advanced pgvector indexing, reranking, evaluation, and observability.

• Fine-grained PostgreSQL row-level security policies for tenant isolation.

• Real payment gateway and subscription billing integration.

• Automated deployment to cloud, private VPC, and air-gapped environments.

• Third-party developer publishing and component certification.

• Advanced analytics, monitoring, alerting, and automated license enforcement.

15. Conclusion

Cortex Hub presents a practical enterprise software solution that combines an AI component marketplace with customization, licensing, customer management, and AI-service infrastructure. The use of real PostgreSQL 18 with Prisma ORM is a central implementation feature because it provides persistent, structured, and reliable storage for the platform's multi-tenant business workflows. The project therefore goes beyond a static frontend demonstration and establishes a foundation for a deployable SaaS product.

16. References

• Cortex Hub repository: https://github.com/Goldypahal/Cortex-Hub

• Project Software Requirements Specification: capstone.md in the Cortex Hub repository.

• PostgreSQL official documentation: https://www.postgresql.org/docs/

• Prisma ORM documentation: https://www.prisma.io/docs/